CSE STUDENT PLACEMENT INTELLIGENCE SYSTEM

Complete Project Specification: Features List, Modules Directory, Sprint Roadmap (V1–V3), User Stories & System Architecture Diagrams

Target Audience:	Computer Science & Engineering Students, Faculty Mentors, TPO Staff
Document Type:	Comprehensive Product Requirement Document (PRD) & Sprint Guide
Project Scope:	Version 1.0 (Core CS & Skill Engine) to Version 3.0 (Advanced Intelligence)
Date & Status:	October 2026 | Approved Development Specification

> **Executive Summary for Project Students**
> This document is designed to give you a complete, step-by-step roadmap for building the CSE Student Placement Intelligence System. Unlike traditional college placement portals that only look at static CGPA or self-reported skills, this system acts as an intelligent decision platform. It evaluates students dynamically based on actual assessments, mock interviews, and project evaluations, maintaining a complete historical log. By reading this document, you will understand every feature, module, sprint, user story, and architectural diagram required to build this application.

1. Comprehensive Features Catalog

The CSE Student Placement Intelligence System is engineered to transform raw assessment data into actionable placement intelligence. Below is the complete list of features implemented across the application, written in student-friendly terms:

- **1. Dynamic Multi-Source Skill Evaluation Engine:** Instead of manually entering fixed skill ratings, the system dynamically calculates a student's current skill score by aggregating historical results from Coding Tests, MCQs, Mock Interviews, Project Reviews, and Trainer Assessments.
- **2. Dedicated Core Computer Science Index:** Core CS subjects (DBMS, SQL, Operating Systems, Computer Networks, Cloud Computing, and Computer Architecture) are tracked as an independent,

permanent V1 metric. This gives placement teams instant visibility into a student's fundamental interview readiness.

- **3. Student 360° Profile & Visual Dashboard:** A unified visual scorecard for every student displaying percentage breakdown across four primary domains: Programming & Problem Solving, Core Computer Science, Software Development, and Employability/Soft Skills.
- **4. Automated Talent Persona Classification:** An intelligent tagging algorithm automatically categorizes students into actionable talent personas such as '🟢 DSA Strong', '🟢 Core CS Strong', '🟢 Development Strong', '🟢 Communication Strong', '🟢 All-Rounder', or '🟡 Needs Targeted Training'.
- **5. Multi-Criteria Candidate Search & Query Engine:** Training & Placement Officers (TPOs) can run complex queries to filter candidates (e.g., 'Find students with DBMS >= 80%, OS >= 80%, Networks >= 80%, and Communication >= 75%' for high-tier technical roles).
- **6. Historical Skill Progress & Growth Tracking:** Maintains full historical integrity for every interview and assessment. Dashboard graphs display skill trajectories over time (e.g., tracking a student's DSA score growth from 58% in January to 86% in June).
- **7. Mock Interview & Structured Feedback Engine:** Interviewers can log qualitative strengths/weaknesses, unanswered questions, recommended training, and select automated recommendations (e.g., DIRECT_REFERRAL_REVIEW, COMPANY_READY, TARGETED_TRAINING).
- **8. Intelligent Company Requirements & Matching Engine:** Companies define specific job roles with required skill scores and percentage weightages (e.g., Java/OOP 20%, DSA 20%, DBMS 10%, Project 15%). The system calculates a customized Match Percentage (e.g., Rahul -> Java Developer Role = 87% Match).
- **9. Placement Drive Lifecycle Management:** End-to-end management of recruitment drives, eligibility verification, student application tracking, interview round progression, and final job offer logging.
- **10. Role-Based Access Control (RBAC) & Audit Log:** Secure access levels tailored for Administrators, Trainers/Interviewers, and Students, complete with audit timestamps for transparent evaluation.

2. System Modules Directory

To make development manageable, the system is partitioned into eight clear, modular components. Each module handles a specific aspect of the system lifecycle:

Module ID	Module Name	Primary Purpose	Key Operations & Outputs
M-01	User Management & Auth	Handles authentication, security, and role-based access.	User signup/login, BCrypt encryption, role assignment (ADMIN, TRAINER, STUDENT).
M-02	Skill Taxonomy & Core CS	Defines the 4 primary skill	Category management,

	Engine	domains and 19 sub-skills.	Core CS subject grouping, score aggregation formulas.
M-03	Assessment & Evaluation Engine	Schedules tests and logs quantitative evaluation scores.	Coding test logging, MCQ score entry, project rubric evaluations, result publishing.
M-04	Mock Interview & Feedback	Captures detailed technical and HR mock interview sessions.	Interviewer feedback capture, rating submission, system recommendation flags.
M-05	Student 360° Profile & Dashboard	Consolidates all student data into a interactive portal.	Student profile generation, dynamic skill scorecard display, persona badge assignment.
M-06	Company Matching Engine	Configures company role requirements and computes match scores.	Job requirement weightage setup, student-role matching algorithm, fit analysis.
M-07	Placement Drive Management	Manages hiring drives, applications, and selection rounds.	Drive creation, eligibility filtering, applicant tracking, final offer logging.
M-08	Analytics & Reporting Engine	Generates intelligence reports and candidate search results.	TPO candidate search queries, skill growth trend charts, intervention lists.

3. Development Roadmap: Sprints & User Stories (V1 to V3)

Development is structured into Agile Sprints spanning three major software versions. This allows students to build the application incrementally, starting from database foundations up to advanced matching algorithms.

Version 1.0: Core Skill Tracking & Assessment Engine (Sprints 1 – 3)

Sprint 1: System Infrastructure, User Auth & Skill Taxonomy

- **US-1.1 (Database Infrastructure):** As a Developer, I want to set up the MySQL database schema and JPA entities so that all core application tables are properly mapped.

- **US-1.2 (Role-Based Auth):** As a User, I want to log in using my credentials and be routed to my role-specific dashboard (Admin, Trainer, Student).
- **US-1.3 (Skill Taxonomy Management):** As an Admin, I want to manage the Skill Taxonomy table pre-populated with 4 domains and 19 skills including Core CS subjects.
- **US-1.4 (Student Profile Setup):** As a Student, I want to view my personal academic profile details (Enrollment No, CGPA, Branch, Batch).

Sprint 2: Assessment Engine, Mock Interviews & Evaluation

- **US-2.1 (Assessment Logging):** As a Trainer, I want to create assessment events (Coding Test, MCQ, Project Review) and upload student scores.
- **US-2.2 (Mock Interview Entry):** As a Trainer, I want to conduct mock interviews and record detailed qualitative feedback (Strengths, Weaknesses, Unanswered Questions).
- **US-2.3 (Multi-Source Skill Rating):** As a Trainer, I want to rate specific skills on a 1–5 scale during assessments, auto-calculating score percentages in skill_ratings.
- **US-2.4 (Evaluation Audit Trail):** As a System, I want to store source_type and source_id for every rating so that skill scores remain verifiable and transparent.

Sprint 3: Dynamic Skill Scoring, Core CS Index & Student 360

- **US-3.1 (Dynamic Score Aggregation):** As a System, I want to aggregate a student's historical skill_ratings into a real-time average percentage per skill.
- **US-3.2 (Core CS Index Display):** As a Student/Trainer, I want to see a dedicated Core CS Readiness indicator aggregating DBMS, OS, Networks, Cloud, and Architecture.
- **US-3.3 (Student 360 Dashboard):** As a Student, I want to view my Student 360° Profile dashboard with progress bars across all 4 skill domains.
- **US-3.4 (Automated Persona Tagging):** As a Placement Team Member, I want the system to auto-assign Talent Persona badges (e.g., 'DSA Strong', 'Core CS Strong').

Version 2.0: Company Requirements & Matching Engine (Sprints 4 – 5)

Sprint 4: Company Requirements & Compatibility Engine

- **US-4.1 (Company Requirement Configuration):** As an Admin, I want to create company profiles and define specific skill requirements with cutoffs and weightages.
- **US-4.2 (Automated Role Matching):** As a Placement Officer, I want the system to compute a weighted Compatibility Match Percentage between a student and a job role.
- **US-4.3 (Advanced Talent Query Search):** As a TPO, I want to run multi-criteria filter queries (e.g., DBMS >= 80% AND OS >= 80% AND Communication >= 75%) to find ideal candidates.

Sprint 5: Placement Drives & Selection Tracking (optional)

- **US-5.1 (Placement Drive Scheduling):** As a TPO, I want to schedule placement drives and automatically notify eligible students based on CGPA and skill thresholds.
- **US-5.2 (Drive Application):** As a Student, I want to view upcoming drives and apply for roles where I meet eligibility requirements.
- **US-5.3 (Selection Round Progression):** As a TPO, I want to update student progression through interview rounds (Shortlisted, Technical Round, HR Round, Selected).

- **US-5.4 (Placement Result Logging):** As an Admin, I want to record final placement offers, accepted packages (CTC in LPA), and offer letters.

Version 3.0: Advanced Analytics, Growth Timeline & Interventions (Sprint 6)

Sprint 6: Skill Growth Timelines & Strategic Reporting

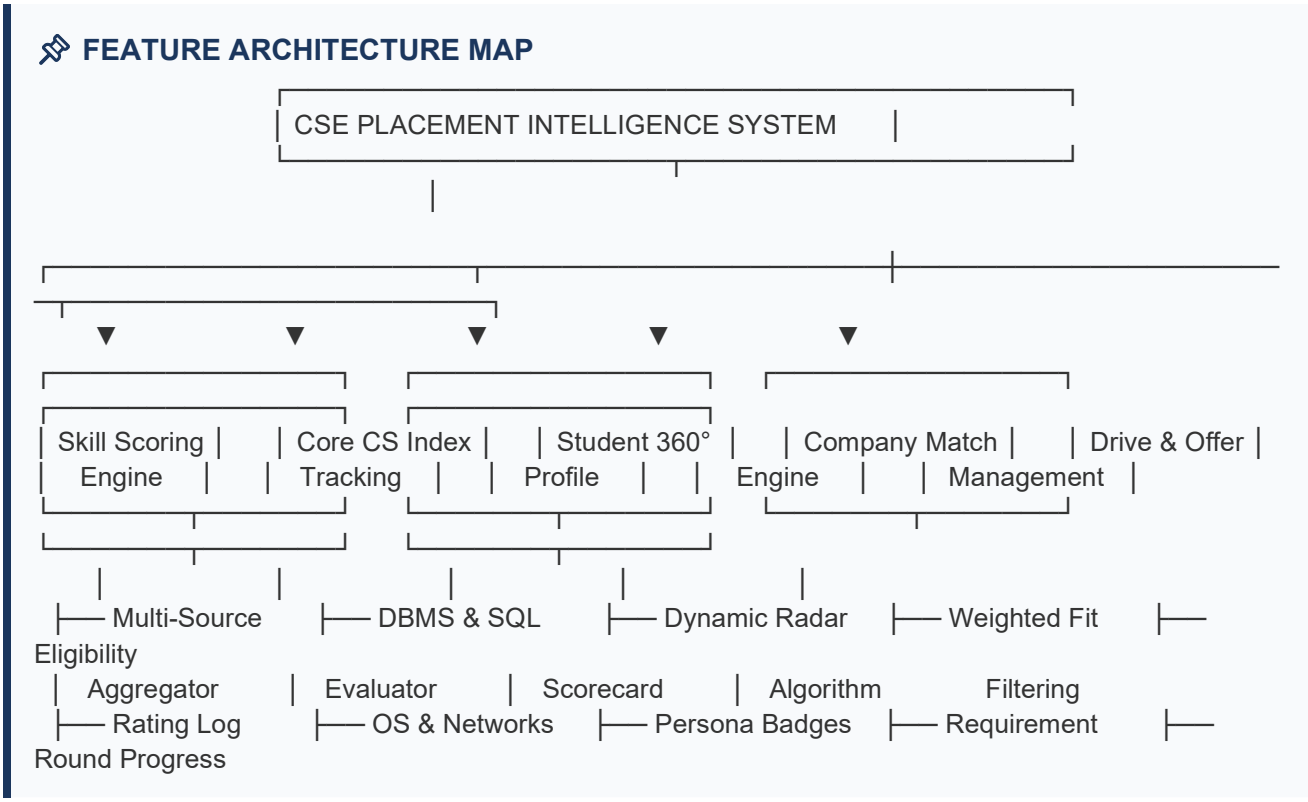
- **US-6.1 (Historical Growth Timeline):** As a Student, I want to see line charts of my skill progression over time to see how my performance has improved across months.
- **US-6.2 (Targeted Training Intervention):** As a Trainer, I want the system to flag students marked 'Needs Targeted Training' and highlight specific weak sub-skills.
- **US-6.3 (Exportable Intelligence Reports):** As a Placement Director, I want to export overall placement readiness reports, batch analytics, and company-wise candidate shortlists.

4. System Diagrams & Architecture

To help students visualize the overall structure, process flow, and technical implementation, three core architectural diagrams are provided below:

Diagram 1: Feature Architecture Map

The following structural map shows how core project features branch out across functional domains:



```
  │ Audit          │ Monitor        │ (DSA/Core CS)  │ Cutoffs          Tracker
  └── Growth Trend   └── Cloud/Arch   └── Progress Log  └── Multi-Criteria  └── Result &
Package
    Calculator       Scorer           History          Talent Search      Logger
```

Diagram 2: End-to-End System Process Flow

This diagram illustrates the step-by-step data pipeline from evaluation event creation down to company job referral:

📌 END-TO-END DATA & PROCESS FLOW PIPELINE

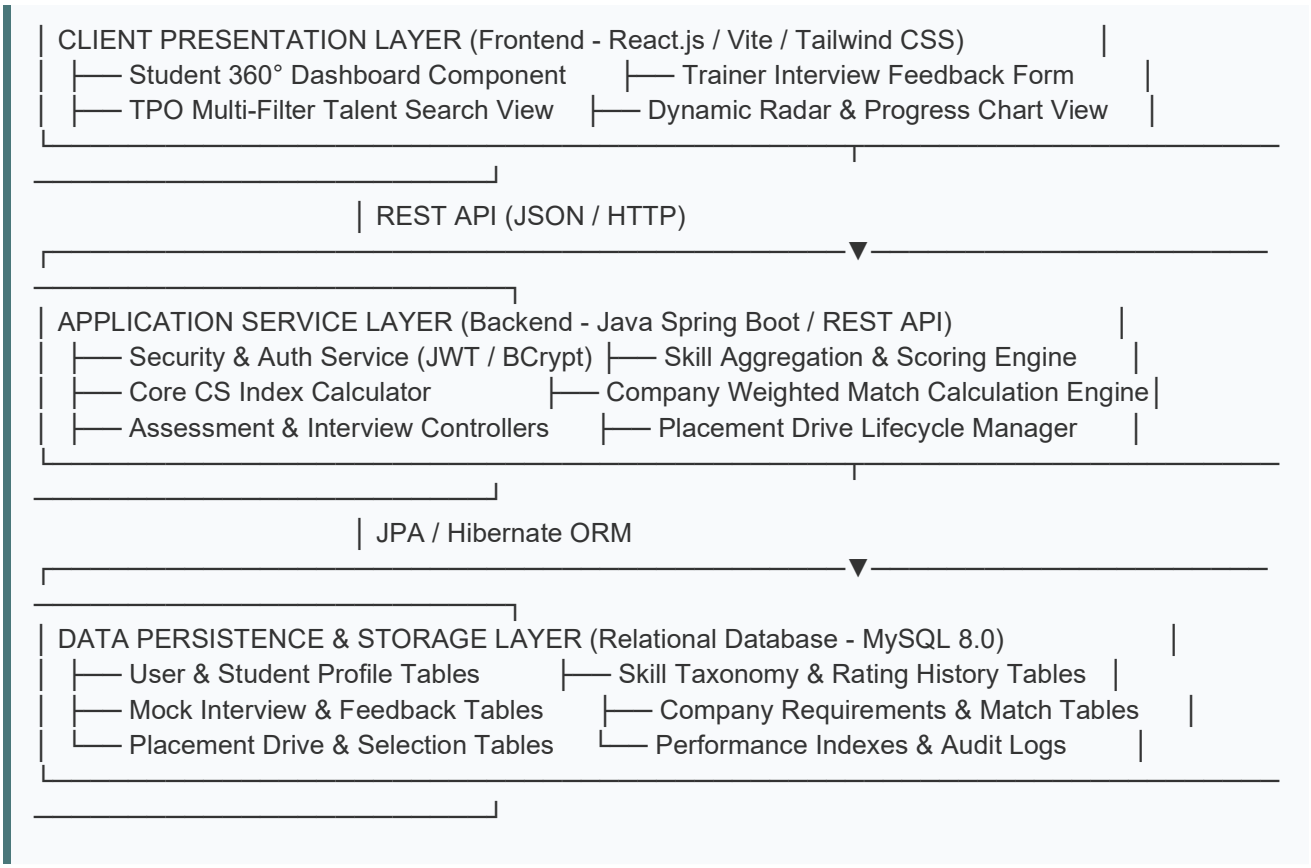
[Step 1: Evaluation Event] ──► Trainer conducts Coding Test / Mock Interview / Project Review
                                  │
                                  ▼
[Step 2: Rating Capture] ──► Detailed skill_ratings logged (Rating 1-5, %, source_type, source_id)
                                  │
                                  ▼
[Step 3: Aggregation Engine]──► System recalculates dynamic skill averages across 4 domains
                                  │
                                  ▼
[Step 4: Profile Update] ──► Student 360° Profile updated & Core CS Readiness Index generated
                                  │
                                  ▼
[Step 5: Persona Tagging] ──► System assigns tags (DSA Strong, Core CS Strong, All-Rounder)
                                  │
                                  ▼
[Step 6: Company Matching] ──► Job requirement weightage executed to calculate Match Percentage (%)
                                  │
                                  ▼
[Step 7: TPO Shortlist] ──► Placement Officer executes multi-criteria search & shortlists students
                                  │
                                  ▼
[Step 8: Drive Selection] ──► Student applies -> Round progression tracked -> Offer & CTC logged

Diagram 3: System Development & Technology Layer Diagram

This diagram details the recommended 3-Tier Technical Architecture for students implementing the stack:

📌 3-TIER TECHNICAL DEVELOPMENT ARCHITECTURE




CLIENT PRESENTATION LAYER (Frontend - React.js / Vite / Tailwind CSS)
Student 360° Dashboard Component
Trainer Interview Feedback Form
TPO Multi-Filter Talent Search View
Dynamic Radar & Progress Chart View
REST API (JSON / HTTP)
APPLICATION SERVICE LAYER (Backend - Java Spring Boot / REST API)
Security & Auth Service (JWT / BCrypt)
Skill Aggregation & Scoring Engine
Core CS Index Calculator
Company Weighted Match Calculation Engine
Assessment & Interview Controllers
Placement Drive Lifecycle Manager
JPA / Hibernate ORM
DATA PERSISTENCE & STORAGE LAYER (Relational Database - MySQL 8.0)
User & Student Profile Tables
Skill Taxonomy & Rating History Tables
Mock Interview & Feedback Tables
Company Requirements & Match Tables
Placement Drive & Selection Tables
Performance Indexes & Audit Logs